

NORTH SEA STATION 63103, United Kingdom

WMO: 995780

Lat: 61.100N

Lon: 1.100E

Elev: 0

StdP: 14.70

Time Zone: 0.00 (EUW)

Period: 94-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	32.7	34.4	17.8	13.5	36.4	20.0	15.0	37.6	50.5	39.8	48.3	40.2	19.7	350	0.921

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	3.4	63.3	59.6	61.6	58.6	60.2	57.3	60.8	62.7	59.6	61.4	58.2	60.0	13.8	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.0	77.2	61.9	58.6	73.5	60.6	57.3	70.0	59.1	26.9	62.9	26.1	61.6	25.2	60.1	68.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 45.2	40.4	35.5	DB	29.5	67.2	1.8	4.3	28.2	70.3	27.1	72.8	26.1	75.2	24.7	78.3	(4)
(5)			WB	27.3	62.5	2.1	3.3	25.8	64.9	24.6	66.8	23.4	68.7	21.9	71.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	48.1	42.7	41.6	42.8	44.5	47.8	51.4	55.7	56.7	54.2	49.2	46.8	43.2	(6)
(7)		DBStd	6.09	3.16	3.70	3.53	3.06	2.74	3.12	2.91	2.77	3.05	3.13	2.91	4.17	(7)
(8)		HDD50	1310	227	236	224	165	77	15	0	0	5	51	98	212	(8)
(9)		HDD65	6173	692	656	689	615	533	407	287	257	323	491	546	677	(9)
(10)		CDD50	614	0	0	0	1	9	58	178	208	132	25	3	0	(10)
(11)		CDD65	1	0	0	0	0	0	0	0	1	0	0	0	0	(11)
(12)		CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0	(12)
(13)		CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)
(14)	Wind	WSAvg	18.6	22.6	21.9	20.9	18.0	15.5	14.8	14.6	14.7	17.6	20.1	21.6	21.7	(14)
(15)	Precipitation	PrecAvg	52.8	6.3	5.2	5.0	2.8	1.9	2.6	3.2	3.9	4.6	5.6	5.5	6.1	(15)
(16)		PrecMax	65.0	9.9	10.1	9.3	4.8	3.3	5.2	6.0	6.4	7.5	8.5	8.7	9.4	(16)
(17)		PrecMin	37.6	3.0	2.3	1.8	1.0	0.9	1.0	1.6	2.6	1.9	2.3	1.7	2.6	(17)
(18)		PrecStd	6.5	1.9	2.1	1.8	0.9	0.6	0.9	0.9	1.0	1.3	1.6	1.5	1.7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.0	50.0	49.5	52.2	55.3	64.6	65.9	67.3	62.9	57.0	52.9	51.2	(19)
(20)			MCWB	47.4	47.6	46.8	48.7	48.7	57.6	61.4	61.7	58.9	53.9	50.2	48.8	(20)
(21)		2%	DB	48.8	48.6	48.6	50.3	53.7	60.3	63.4	63.0	60.2	55.7	51.9	50.4	(21)
(22)			MCWB	46.0	46.1	46.1	47.6	49.2	56.3	60.3	59.7	57.6	52.8	49.6	47.9	(22)
(23)		5%	DB	48.0	47.8	48.0	49.2	52.4	56.7	61.5	61.6	59.1	54.5	51.2	49.8	(23)
(24)			MCWB	45.0	45.1	45.6	46.7	48.8	53.9	59.1	58.5	56.9	51.4	48.7	47.1	(24)
(25)		10%	DB	47.3	47.1	47.3	48.3	51.3	54.9	59.8	60.3	58.1	53.1	50.5	49.0	(25)
(26)			MCWB	44.2	44.2	44.8	46.0	48.2	52.2	57.5	57.1	55.8	49.9	47.9	45.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.5	48.8	48.5	50.7	51.9	59.4	63.0	62.3	60.0	55.5	51.7	50.4	(27)
(28)			MCDWB	49.4	49.2	48.8	50.8	53.1	62.8	64.4	64.6	61.6	56.7	52.3	50.7	(28)
(29)		2%	WB	47.4	47.6	47.5	48.6	51.0	56.9	61.1	60.9	58.7	53.8	50.7	49.3	(29)
(30)			MCDWB	48.5	48.3	48.1	49.0	52.3	60.7	62.7	63.0	60.0	55.7	51.6	50.1	(30)
(31)		5%	WB	46.2	46.7	46.6	47.8	50.0	55.3	59.9	59.6	57.6	52.4	49.8	47.9	(31)
(32)			MCDWB	47.7	47.7	47.6	48.5	51.5	58.1	61.5	61.6	58.7	54.0	50.9	49.3	(32)
(33)		10%	WB	45.0	45.4	45.6	46.8	49.0	53.5	58.2	58.1	56.8	51.0	48.7	46.3	(33)
(34)			MCDWB	47.0	46.8	46.9	48.0	50.9	55.1	59.9	59.9	57.9	52.8	50.2	48.4	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	4.8	5.0	4.6	4.1	3.6	3.4	3.1	3.4	3.7	3.8	4.1	4.7	(35)
(36)			MCDBR	5.0	5.0	4.3	4.7	4.3	5.0	4.7	4.4	4.5	3.8	4.4	4.7	(36)
(37)			MCWBR	6.1	5.7	4.8	4.2	3.3	3.6	3.6	3.8	3.7	4.4	5.6	6.6	(37)
(38)		5% WB	MCDBR	5.2	4.6	4.4	4.5	3.5	4.9	3.9	4.2	3.8	3.5	4.2	5.1	(38)
(39)			MCWBR	6.7	6.3	5.2	4.5	3.1	3.8	3.7	4.0	3.6	4.7	5.9	7.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.338	0.357	0.370	0.382	0.372	0.364	0.382	0.371	0.366	0.348	0.324	0.383	(40)	
(41)		taud	2.062	2.181	2.261	2.301	2.401	2.464	2.432	2.462	2.423	2.303	2.126	2.284	(41)	
(42)		Ebn at Noon	108	188	233	256	270	274	265	259	236	192	114	68	(42)	
(43)		Edh at Noon	14	23	29	34	33	31	32	29	25	20	13	9	(43)	
(44)	All-Sky Solar Radiation	RadAvg	61	216	573	1114	1553	1583	1437	1152	699	329	95	38	(44)	
(45)		RadStd	8	26	60	87	111	170	131	108	55	40	12	2	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (23 neighbors)	N/A	+1.17	+1.12	N/A	N/A	N/A	-184	-241	N/A	N/A	

Nomenclature: See separate page